

Visualization Index

Complete catalog of every generated figure in the project, organised by analysis stage. For detailed data-flow traces and downstream usage see [figure_trace_guide.md](#) (figure_trace_guide.md).

Quick Reference

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Demand & EDA	10	demand_modeling.py , 02_eda_spatiotemporal.ipynb
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Publication Quality	5	generate_publication_figures.py
Dashboard	1	generate_summary_dashboard.py
Total	~60	

1 Demand & EDA

Figure	Description	Script / Notebook
fig_hourly_demand.png	24-hour crash demand profile	notebooks/ 02_eda_spatiotemporal.ipynb
fig_daily_demand.png	Day-of-week crash volume	notebooks/ 02_eda_spatiotemporal.ipynb
fig_temporal_trends.png	Long-term crash trends 2012-2026	notebooks/ 02_eda_spatiotemporal.ipynb
fig_crash_heatmap.png	Spatial heatmap of crash incidents	notebooks/ 02_eda_spatiotemporal.ipynb
fig_precinct_density.png	Precinct demand density choropleth	notebooks/ 02_eda_spatiotemporal.ipynb
fig_firehouses_map.png	Map of 48 Manhattan firehouses	notebooks/ 02_eda_spatiotemporal.ipynb
fig_cbd_comparison.png	CBD robustness comparison	notebooks/ 02_eda_spatiotemporal.ipynb
fig_precinct_demand.png	Precinct demand bar chart (original)	scripts/demand_modeling.py
pre- cinct_demand_rates_improved .png	Improved bar chart with legend, annotations	scripts/gener- ate_precinct_demand_visuali- zations.py
precinct_demand_heatmap.png	Spatial heatmap of precinct demand	scripts/gener- ate_precinct_demand_visuali- zations.py
fig_demand_model_fit.png	NHPP model fit diagnostics	scripts/demand_modeling.py
fig_hourly_rates.png	CBD vs non-CBD lambda factors	scripts/demand_modeling.py

2 Service & Travel Models

Figure	Description	Script / Notebook
<code>distance_matrix_heatmap.png</code>	Haversine distance matrix heatmap	<code>notebooks/04_service_travel_proxy.ipynb</code>
<code>travel_time_by_tod.png</code>	Travel time by time-of-day band	<code>notebooks/04_service_travel_proxy.ipynb</code>
<code>tod_speed_factors.png</code>	24-hour speed factor profile	<code>notebooks/04_service_travel_proxy.ipynb</code>
<code>service_time_distribution.png</code>	LogNormal service time distribution	<code>notebooks/04_service_travel_proxy.ipynb</code>
<code>nhpp_arrivals_demo.png</code>	NHPP arrival generation demo	<code>notebooks/04_service_travel_proxy.ipynb</code>

3 Optimization

Figure	Description	Script / Notebook
<code>fig_policy_comparison.png</code>	Policy comparison (P0, P1, P2)	<code>scripts/run_optimization_comparison.py</code>
<code>fig_tradeoff_curve.png</code>	RT vs coverage trade-off	<code>scripts/run_optimization_comparison.py</code>
<code>opt_allocation_comparison.png</code>	Allocation maps by model	<code>scripts/run_optimization_comparison.py</code>
<code>opt_inputs.png</code>	Optimization inputs visualization	<code>scripts/run_optimization_comparison.py</code>
<code>opt_sensitivity.png</code>	Objective vs fleet size	<code>scripts/run_optimization_comparison.py</code>

4 Simulation Experiments

Figure	Description	Script / Notebook
<code>exp1_policy_comparison.png</code>	Experiment 1: Policy box plots	<code>notebooks/07_production_results.ipynb</code>
<code>exp2_fleet_sensitivity.png</code>	Experiment 2: RT vs fleet size	<code>notebooks/07_production_results.ipynb</code>
<code>exp3_demand_sensitivity.png</code>	Experiment 3: RT vs demand multiplier	<code>notebooks/07_production_results.ipynb</code>
<code>exp4_service_robustness.png</code>	Experiment 4: RT vs service time	<code>notebooks/07_production_results.ipynb</code>

5 Verification & Validation

Figure	Description	Script / Notebook
<code>verification_toy_timeline.png</code>	Toy example timeline	<code>notebooks/06_simulation_debug.py</code>
<code>validation_p0_vs_p2.png</code>	Pilot 1: P0 vs P2	<code>notebooks/06_simulation_debug.py</code>
<code>validation_sensitivity_K.png</code>	Pilot 2: RT vs K	<code>notebooks/06_simulation_debug.py</code>
<code>validation_sensitivity_demand.png</code>	Pilot 3: RT vs demand	<code>notebooks/06_simulation_debug.py</code>

6 Production V2 & Fleet

Figure	Description	Script / Notebook
<code>fleet_sensitivity_v2_dual.png</code>	Dual-panel fleet sensitivity	<code>scripts/run_production_v2.py</code>
<code>response_time_distribution_by_policy.png</code>	RT distribution by policy	Production pipeline
<code>policy_comparison_panel_K20_cap2.png</code>	Staging map panel (P0/P1/P2)	Production pipeline

7 CBD Analysis

Figure	Description	Script / Notebook
<code>cbd_heatmap.png</code>	CBD crash density + fire-houses	<code>notebooks/09_cbd_analysis.ipynb</code>
<code>cbd_response_comparison.png</code>	CBD stress RT comparison	<code>notebooks/09_cbd_analysis.ipynb</code>
<code>cbd_scenario_comparison.png</code>	CBD vs Manhattan scenarios	<code>notebooks/09_cbd_analysis.ipynb</code>
<code>cbd_robustness_enhanced.png</code>	Enhanced CBD robustness	Production pipeline
<code>cbd_equity_tradeoff_summary.png</code>	CBD equity-efficiency trade-off	<code>scripts/run_cbd_focused_optimization.py</code>

8 Queue Analysis

Figure	Description	Script / Notebook
<code>queue_comparison_by_policy.png</code>	Queue metrics by policy	<code>scripts/analyze_queue_metrics.py</code>
<code>queue_heatmap.png</code>	Queue length heatmap	<code>scripts/analyze_queue_metrics.py</code>
<code>queue_vs_demand.png</code>	Queue vs demand multiplier	<code>scripts/analyze_queue_metrics.py</code>
<code>queue_vs_fleet_size.png</code>	Queue vs fleet size	<code>scripts/analyze_queue_metrics.py</code>

9 Seasonal Analysis

Figure	Description	Script / Notebook
<code>seasonal_patterns.png</code>	Seasonal variation analysis	<code>scripts/analyze_seasonal_patterns.py</code>
<code>seasonal_decomposition.png</code>	Seasonal decomposition	<code>scripts/analyze_seasonal_patterns.py</code>
<code>seasonal_heatmap.png</code>	Month x DoW heatmap	<code>scripts/analyze_seasonal_patterns.py</code>

10 Capacity Sensitivity

Figure	Description	Script / Notebook
capa-city_sensitivity_heatmap.png	RT by policy and capacity	scripts/generate_capacity_sensitivity_heatmap.py
fire-house_capacity_analysis.png	Capacity analysis detail	scripts/capacity_sensitivity_analysis.py

11 Distance Comparison

Figure	Description	Script / Notebook
distance_comparison/distance_matrices_heatmap.png	Side-by-side distance heatmaps	scripts/run_distance_comparison_experiment.py
distance_comparison/distance_scatter.png	Haversine vs Manhattan scatter	scripts/run_distance_comparison_experiment.py
distance_comparison/distance_comparison_bar.png	Performance bar chart	scripts/run_distance_comparison_experiment.py
distance_comparison/distance_comparison_boxplot.png	Replication box plots	scripts/run_distance_comparison_experiment.py

12 P0 Spatial Baseline

Figure	Description	Script / Notebook
p0_spatial_map.png	P0-spatial firehouse selections	scripts/p0_spatial_analysis.py
p0_spatial_metrics.png	Stratification comparison metrics	scripts/p0_spatial_analysis.py
p0_spatial_north_south.png	North-south distribution	scripts/p0_spatial_analysis.py

13 Publication-Quality Figures

Figure	Description	Script / Notebook
<code>pub_fig1_policy_comparison.png</code>	Final Figure 1	<code>scripts/generate_publication_figures.py</code>
<code>pub_fig2_fleet_sensitivity.png</code>	Final Figure 2	<code>scripts/generate_publication_figures.py</code>
<code>pub_fig3_demand_robustness.png</code>	Final Figure 3	<code>scripts/generate_publication_figures.py</code>
<code>pub_fig4_service_sensitivity.png</code>	Final Figure 4	<code>scripts/generate_publication_figures.py</code>
<code>pub_fig5_performance_heatmap.png</code>	Final Figure 5	<code>scripts/generate_publication_figures.py</code>

14 Dashboard

Figure	Description	Script / Notebook
<code>project_summary_dashboard.png</code>	Full project summary	<code>scripts/generate_summary_dashboard.py</code>

Regeneration

All figures can be regenerated from source data. See [figure_trace_guide.md](#) (`figure_trace_guide.md`) for the full list of regeneration commands.